

Thm. Every finite-dimensional Inner Product space has an orthonormal basis.

Proof. Let V be an Inner Product space not necessary finite dimensional.

Given a linearly independent subset $S = \{\beta_1, \dots, \beta_n\} \subset V$,

find an orthonormal set S' such that $\text{span } S = \text{span } S'$.

Take $\alpha_1 = \beta_1$ And, take α_2 as

$$\alpha_2 = \beta_2 - \underbrace{\frac{\langle \beta_2, \alpha_1 \rangle}{\|\alpha_1\|^2}}_{=\text{orth. proj. of } \beta_2 \text{ on } \langle \alpha_1 \rangle} \cdot \alpha_1$$

$$\alpha_3 = \beta_3 - \frac{\langle \beta_3, \alpha_1 \rangle}{\|\alpha_1\|^2} \cdot \alpha_1 - \frac{\langle \beta_3, \alpha_2 \rangle}{\|\alpha_2\|^2} \cdot \alpha_2$$

$\vdots$